§3.11.

P1. $A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & 1 \\ -3 & 1 & -1 \end{bmatrix}$

$$P(\lambda) = \det(A-\lambda I) = \det\begin{bmatrix} 1-\lambda & -1 & -1 \\ 1 & 3 & 1 \\ -3 & 1 & -1 \end{bmatrix}$$

$$= -(\lambda+2)(\lambda-2)(\lambda-3) = 0$$

$$\Rightarrow \lambda_1 = -2,\ \lambda_2 = 2,\ \lambda_3 = 3$$

i) $\lambda_1 = -2$

$$(A+2I)V = 0 \Rightarrow \begin{bmatrix} 3 & -1 & -1 \\ 1 & 5 & 1 \\ -3 & 1 & 1 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow V = c\begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$$

$$\Rightarrow x^1(t) = e^{-2t}\begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$$

i) $\lambda_2 = 2$

$$(A-2I)V = 0 \Rightarrow \begin{bmatrix} -1 & -1 & -1 \\ 1 & 1 & 1 \\ -3 & 1 & -3 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = 0$$

$$\Rightarrow V = c\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\Rightarrow x^2(t) = e^{2t}\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

iii) $\lambda_3 = 3$

$$(A-3I)V = 0 \Rightarrow \begin{bmatrix} -2 & -1 & -1 \\ 1 & 0 & 1 \\ -3 & 1 & -4 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow V = c\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow x^3(t) = e^{3t}\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \Phi(t) = \begin{bmatrix} e^{-2t} & e^{2t} & -e^{3t} \\ -e^{-2t} & 0 & e^{3t} \\ 4e^{-2t} & -e^{2t} & e^{3t} \end{bmatrix}$$

$$[\Phi(0)]^{-1} = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 0 & 1 \\ 4 & -1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{5} & 0 & \frac{1}{5} \\ 1 & 1 & 0 \\ \frac{1}{5} & 1 & \frac{1}{5} \end{bmatrix}$$

$$\Rightarrow e^{At} = \Phi(t)[\Phi(0)]^{-1} = \frac{1}{5}\begin{bmatrix} e^{-2t}+5e^{2t}-e^{3t} & 5e^{2t}-5e^{3t} & e^{-2t}-e^{3t} \\ -e^{-2t}+e^{3t} & 5e^{3t} & -e^{-2t}+e^{3t} \\ 4e^{-2t}-5e^{2t}+e^{3t} & -5e^{2t}+5e^{3t} & 4e^{-2t}+e^{3t} \end{bmatrix}$$

P3. $A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & -2 & 0 \end{pmatrix}$

$$P(\lambda) = \det(A-\lambda I) = \det\begin{pmatrix} 2-\lambda & 0 & 1 \\ 1 & -\lambda & 1 \\ 1 & -2 & -\lambda \end{pmatrix}$$

$$= -(\lambda-2)(\lambda^2+1) = 0$$

$$\Rightarrow \lambda_1 = 2,\ \lambda_2 = i,\ \lambda_3 = -i$$

i) $\lambda_1 = 2$

$$(A-2I)V = 0 \Rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 1 & -2 & 1 \\ 1 & -2 & -2 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow V = c\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \Rightarrow x^1(t) = e^{2t}\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

ii) $\lambda_2 = i$

$$(A-iI)V = 0 \Rightarrow \begin{bmatrix} 2-i & 0 & 1 \\ 1 & -i & 1 \\ 1 & -2 & -i \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow V = c\begin{bmatrix} 1 \\ 1+i \\ -2+i \end{bmatrix} \Rightarrow \tilde{x}(t) = e^{it}\begin{bmatrix} 1 \\ 1+i \\ -2+i \end{bmatrix}$$

$$\tilde{x}(t) = (\cos t + i\sin t)\begin{bmatrix} 1 \\ 1+i \\ -2+i \end{bmatrix} = \begin{bmatrix} \cos t + i\sin t \\ \cos t - \sin t + i(\cos t + \sin t) \\ -2\cos t - \sin t + i(\cos t - 2\sin t) \end{bmatrix}$$

$$\Rightarrow x^2(t) = Re(\tilde{x}(t)) = \begin{bmatrix} \cos t \\ \cos t - \sin t \\ -2\cos t - \sin t \end{bmatrix}$$

$$x^3(t) = Im(\tilde{x}(t)) = \begin{bmatrix} \sin t \\ \cos t + \sin t \\ \cos t - 2\sin t \end{bmatrix}$$

$$\Rightarrow \Phi(t) = \begin{bmatrix} 2e^{2t} & \cos t & \sin t \\ e^{2t} & \cos t - \sin t & \cos t + \sin t \\ 0 & -2\cos t - \sin t & \cos t - 2\sin t \end{bmatrix}$$

$$[\Phi(0)]^{-1} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & -2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{3}{5} & -\frac{1}{5} & \frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{bmatrix}$$

$$\Rightarrow e^{At} = \Phi(t)[\Phi(0)]^{-1}$$

$$= \frac{1}{5}\begin{bmatrix} 6e^{2t}-\cos t-2\sin t & -2e^{2t}+2\cos t+4\sin t & 2e^{2t}-2\cos t+\sin t \\ 3e^{2t}-3\cos t-\sin t & -e^{2t}+6\cos t+2\sin t & e^{2t}-\cos t+3\sin t \\ 5\sin t & -10\sin t & 5\cos t \end{bmatrix}$$

P7. $\frac{de^{At}}{dt} = Ae^{At}$

$$\Rightarrow Ae^{At} = \begin{bmatrix} 4e^{2t}-e^{t} & 2e^{2t}-e^{t} & e^{t}-2e^{2t} \\ 2e^{2t}-e^{t} & 4e^{2t}-e^{t} & e^{t}-2e^{2t} \\ 6e^{2t}-3e^{t} & 6e^{2t}-3e^{t} & 3e^{t}-4e^{2t} \end{bmatrix}$$

setting $t=0$, we get

$$A\underset{I}{\underset{\parallel}{e^0}} = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 3 & 3 & -1 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 3 & 3 & -1 \end{bmatrix}$$

P8. No. since $\begin{bmatrix} e^{t} \\ e^{t} \\ 2e^{t} \end{bmatrix} + 2\begin{bmatrix} e^{-t} \\ -e^{-t} \\ e^{-t} \end{bmatrix} = \begin{bmatrix} e^{t}+2e^{-t} \\ e^{t}-2e^{-t} \\ 2(e^{t}+e^{-t}) \end{bmatrix}$, i.e. the columns are linearly dependent.

P9. Yes.

$$\frac{d\,\underline{X}(t)}{dt} = A\underline{X}(t) = \begin{bmatrix} 10\sin 2t & -10\cos 2t & 6e^{2t} \\ 4\sin 2t - 4\cos 2t & -4\sin 2t - 4\cos 2t & 0 \\ -2\sin 2t & 2\cos 2t & 2e^{2t} \end{bmatrix}$$

setting $t=0$ gives us

$$A\underline{X}(0) = \begin{bmatrix} 0 & -10 & 6 \\ -4 & -4 & 0 \\ 0 & 2 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & -10 & 6 \\ -4 & -4 & 0 \\ 0 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} -5 & 0 & 3 \\ -2 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} 2 & -5 & 0 \\ 1 & -2 & -3 \\ 0 & 1 & 2 \end{bmatrix}$$